# Patrick Thangarajah

Simulation & FEA | Testing & Validation | Mechanical Design | Prototyping

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## Summary

Recent MASc graduate in mechanical engineering with practical experience designing, building, and testing hardware systems, including sensor-integrated test setups and control circuitry. Brings simulation, DOE, and data analysis skills to support designs from concept through validation.

## Skills

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| **Design & Prototyping** | SolidWorks, Onshape, 3D Printing, Lathe, Mill, Bandsaw, Drill Press |
| **Controls & Instrumentation** | Arduino, Sensors, Motors, Data Acquisition, Circuit Design, Altium, LTspice |
| **Testing & Analysis** | DOE, JMP, Minitab, MATLAB, Microsoft Excel, Troubleshooting |
| **Simulation & Programming** | FEA, LS-DYNA, LS-PrePost, Python, C/C++ |

## Education

**Master of Applied Science in Mechanical and Mechatronics Engineering** — *Waterloo, ON, Canada*

University of Waterloo — *Sep 2023 - Jul 2026*

- Research in occupant safety and biomechanics

**Bachelor of Applied Science in Mechanical Engineering, Co-op Program** — *Vancouver, BC, Canada*

University of British Columbia — *Sep 2018 - May 2023*

- Biomechanics and Medical Devices Option

## Experience

**Graduate Research Student - Impact Mechanics and Material Characterization Group** — *Waterloo, ON, Canada*

University of Waterloo — *Sep 2023 - Jul 2026*

- Ran a 104-configuration LS-DYNA DOE and used JMP to quantify geometry effects on occupant response, identifying a configuration with 16% lower peak chest deflection
- Validated simulations against physical tests across two configurations and 10 response channels, achieving a 0.71 average ISO 18571 correlation score
- Troubleshot model-to-test discrepancies in positioning, restraint fit, and response timing to improve agreement with physical tests
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)

**Teaching Assistant - MTE 120: Circuits** — *Waterloo, ON, Canada*

University of Waterloo — *May 2024 - Aug 2024*

- Supported laboratory instruction for a 100+ student circuits course, troubleshooting hardware and providing feedback on lab work and exams

**R&D Engineer (Part-time)** — *Remote*

Echoscope — *May 2023 - Sep 2023*

- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functionality testing; all five were approved and used by the client

**Summer Research Student** — *Vancouver, BC, Canada (Remote)*

Centre for Heart Lung Innovation — *May 2022 - Aug 2022*

- Built a Jupyter Notebook transfer-learning script to diagnose diabetic kidney disease using an emerging imaging platform
- Improved diagnostic accuracy by 10% through model optimization, training refinement, and comparison with existing tissue-slide images

**Mechanical Engineering Intern** — *Airdrie, AB, Canada*

GN Corporation — *Jan 2022 - Apr 2022*

- Developed standardized work instructions and SOPs by reviewing engineering drawings, customer specifications, and existing manufacturing processes
- Reduced fiberglass cutting waste by 50% through process analysis
- Proposed a material-handling process improvement projected to reduce pick-and-place time by 50%

**Engineering Intern** — *Victoria, BC, Canada*

Breathe Global — *Jul 2021 - Aug 2021*

- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components
- Expanded the device's achievable breathing range from 70% to 88% of target

## Projects

**Effect of Seat Foam Properties on Injury Response in Rear Impact (ME 725 Course Project)** — *Waterloo, ON, Canada*

University of Waterloo — *Jan 2024 - Apr 2024*

- Verified 5 polyurethane foam material models in LS-DYNA against input stress-strain response using single-element simulations, then implemented them in a rear-impact sled model with a Hybrid III ATD to calculate HIC15 and Nij injury metrics
- Determined that all 5 foams met the regulatory head injury limit but exceeded the regulatory neck injury limit at high speed, and that the softest foam reduced head injury risk by roughly 18–20% relative to the stiffest

**Simulation of Cervical Fractures in American Football (ME 720 Course Project)** *Waterloo, ON, Canada*

University of Waterloo *Sep 2023 - Dec 2023*

- Simulated head-first helmet impacts across three velocities (3.4–7.0 m/s) and three neck angles at the lowest speed condition using Hybrid III and GHBMC human body models in LS-DYNA, then evaluated fracture force thresholds and injury location against peer-reviewed literature
- Discovered that predicted peak neck forces (2.26–2.34 kN) fell within the range reported across two cadaveric studies, and that the Neck Injury Criterion exceeded its injury-risk threshold in only 1 of 5 configurations despite fracture occurring in all 5

**Soft Distensible Bladder Model - Capstone Project** *Vancouver, BC, Canada*

University of British Columbia *Sep 2022 - Apr 2023*

- Prototyped bladder phantoms using silicone and agar, comparing pressure-volume response with literature to guide material selection
- Designed and built the control and data-acquisition interface for a pressurized soft-bladder test system, enabling real-time pressure/volume monitoring and automated filling/voiding cycles
- Validated system performance using integrated pressure sensing and controls, demonstrating stable and repeatable operation up to 4.75 kPa internal pressure

**Payload Team Member** *Vancouver, BC, Canada*

UBC Unmanned Aircraft Systems (Student Design Team) *Sep 2019 - May 2021*

- Designed and prototyped a payload drop system and rover in Onshape and SolidWorks, iterating through 3D printed proof-of-concept builds
- Reduced payload system weight by 50% by repurposing components from previous designs during concept development

**Volunteer Researcher - Vacuum Injection Molding End-Effector (VIMEE) Gripper** *Vancouver, BC, Canada*

University of British Columbia Collaborative Advanced Robotics and Intelligent Systems (CARIS) Lab *Jan 2022 - Apr 2023*

- Programmed Arduino-controlled LED signal patterns for pleat loading height and force feedback and contributed to user testing of the complete system

## Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.